ResiDual for Audio: Spectral Reweighting of Residual Streams in CLAP Models

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Abstract

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1. Method

Our approach consists of two components: (1) a systematic analysis of the residual stream of CLAP's audio encoder to characterise the behaviour of individual attention heads, and (2) the design of spectral reweighting strategies—collectively referred to as **ResiDual**—that leverage this characterisation to improve downstream performance. This section describes the analysis framework; implementation details of the ResiDual adaptation are deferred to Section 1.4. Full architectural details and notation are collected in Appendix A.

1.1. Residual Stream Decomposition

We analyse the HTS-AT audio encoder (**?**) inside CLAP (**?**). HTS-AT is a hierarchical Swin-Transformer with four stages of block depths $[2, 2, 6, 2]$ and window size $w = 8$. The number of attention heads doubles at each stage, $H_\ell = 4 \cdot 2^\ell$, while the per-head dimension $d_h = 24$ stays constant, so the total embedding dimension $D_\ell = H_\ell \cdot d_h$ also doubles at each transition. Full stage parameters are given in Table 2 in Appendix A.

Within-stage residual structure. Inside each stage ℓ, both the attention and MLP sub-layers write additively to the residual stream. Following the pre-norm formulation (**?**), the stream after block b is:

$$\mathbf{Z}^{(\ell,b)} = \mathbf{Z}^{(\ell,0)} + \sum_{b'=1}^{b} \mathbf{A}_{\ell,b'} + \sum_{b'=1}^{b} \mathbf{M}_{\ell,b'}, \quad \in \mathbb{R}^{S_\ell^2 \times D_\ell}, \tag{1}$$

where $\mathbf{Z}^{(\ell,0)}$ is the stage input, $\mathbf{A}_{\ell,b}$ is the W-MSA output, $\mathbf{M}_{\ell,b}$ is the MLP output, and S_ℓ is the spatial side length of the token grid at stage ℓ.

This decomposition holds strictly *within* a stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension ($S_\ell \to S_\ell/2$, $D_\ell \to 2D_\ell$), breaking any global additive structure across the full network, unlike in isotropic vision transformers (**?**).

Per-head output of the attention layer. The W-MSA output $\mathbf{A}_{\ell,b}$ is produced by concatenating the outputs of H_ℓ independent attention heads and projecting through a shared output matrix $W^O_{\ell,b}$. At block (ℓ, b), the raw output of head h is (see Appendix A.2 for derivation):

$$\mathbf{H}_{\ell,b,h} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right) \mathbf{V}_h \ \in \ \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{2}$$

where $N_w^\ell = S_\ell^2/M$ is the number of attention windows, $M = 64$ tokens per window, and $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head h at block (ℓ, b) (shared across windows, independent per head and per block). All H_ℓ head outputs are concatenated and projected to give the W-MSA output (see Appendix A.3 for the full decomposition):

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \ \in \ \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}. \tag{3}$$

We analyse the raw head outputs $\mathbf{H}_{\ell,b,h}$ directly, *before* the output projection $W^O_{\ell,b}$. Each $\mathbf{H}_{\ell,b,h}$ lives in the per-head space $\mathbb{R}^{d_h}$, which is constant across all stages and blocks. This choice avoids entangling information across heads introduced by the shared projection, and enables uniform cross-stage comparisons of head geometry throughout the network. A complementary analysis of the projected contributions $\widehat{\mathbf{H}}_{\ell,b,h}$—which characterise how each head shapes the residual stream—is provided in Appendix B.

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1.2. Spatial Aggregation and Dataset Representations

Each raw head output $\mathbf{H}_{\ell,b,h}$ is a three-dimensional tensor indexed by window, token position, and feature dimension. To obtain a single fixed-size vector per audio sample, we mean-pool over all windows and token positions:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell \, M} \sum_{i=1}^{N_w^\ell} \sum_{j=1}^{M} \mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \quad (4)$$

This yields one vector $\mathbf{r}_{\ell,b,h}$ per audio sample, per head h, at every block (ℓ, b) of the network. Stacking these vectors across the n samples in the dataset gives the matrix

$$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}, \quad (5)$$

which is the primary object of our analysis. The total number of such matrices across the model is $H_{\text{tot}} = 184$ (one per head per block; see Table 2).

Since $d_h = 24$ is constant across all stages, the ambient dimension of $\mathbf{R}_{\ell,b,h}$ is uniform throughout the network, enabling direct cross-stage comparisons of all dimensionality metrics without correction for a varying ceiling.

Approximation scope and token-level reweighting. The mean-pooled representation $\mathbf{r}_{\ell,b,h}$ is used exclusively for the dimensionality analysis reported in this section. Pooling over all $N_w^\ell \cdot M$ spatial positions collapses the intra-sample token variance, retaining only the inter-sample variance that drives the PCA basis estimation. This choice is motivated by computational constraints: constructing the full token-level matrix $\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n\, N_w^\ell M \times d_h}$ for all 184 heads simultaneously would exceed available memory (considering the all linear and non linear estimators). Despite this approximation, the pooled statistics provide a faithful characterisation of the dominant geometric structure of each head's subspace, since the inter-sample variance captures the directions along which heads discriminate between audio inputs—which is precisely the quantity relevant to understanding head specialisation.

The spectral reweighting in ResiDual, as specified in the next sections, operates instead at full spatial resolution.This token-level granularity allows ResiDual to modulate the geometry of individual spatial positions rather than acting on a global summary, making the intervention more targeted without sacrificing the precision of the basis estimation.

Hook-based extraction. To obtain $\mathbf{H}_{\ell,b,h}$ without modifying the model, we register forward hooks on the `WindowAttention` module of each block (ℓ, b). The hook intercepts the tensor `(attn @ v)` immediately after the per-head attention computation of Eq. (2), yielding the raw outputs $\mathbf{H}_{\ell,b,h}$ of shape $(N_w^\ell,\, M,\, d_h)$. No further projection is applied: the hook output is spatially pooled directly via Eq. (4) to produce $\mathbf{r}_{\ell,b,h}$.

Dataset sampling. We extract representations from four audio classification benchmarks:

- **ESC-50** (**?**): 50 environmental sound classes, 2,000 clips (5 s, 44.1 kHz).
- **TinySOL** (**?**): 14 orchestral instrument classes with varied articulations, 2,071 monophonic samples (1–16 s, 44.1 kHz).
- **VocalSound** (**?**): 6 non-speech vocal categories, stratified subset of 1,200 samples.
- **IRMAS** (**?**): 11 musical instrument classes, annotated audio excerpts with predominant instrument.

Audio preprocessing follows the CLAP standard pipeline: 64-band log-mel spectrogram ($f_{\min} = 50\,\text{Hz}$, $f_{\max} = 8000\,\text{Hz}$, FFT window 1024, hop 320), padded or truncated to 7 seconds at 44.1 kHz.

1.3. Intrinsic Dimensionality Analysis

To characterize the effective complexity of the projected head representations $\{\mathbf{r}_{\ell,b,h}^{(i)}\}_{i=1}^{n} \subset \mathbb{R}^{1024}$, we employ a battery of linear and nonlinear dimensionality estimators.

1.3.1. Linear Estimators

PCA-based dimensionality. For each head, let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ stack all aggregated representations. We compute the covariance matrix $\mathbf{C}_{\ell,b,h} = \frac{1}{n-1}\mathbf{R}^\top \mathbf{R}$ and obtain ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$. Linear intrinsic dimensionality is:

$$d_{\text{PCA}}(\alpha) = \arg\min_k \left\{ \frac{\sum_{i=1}^{k} \lambda_i}{\sum_i \lambda_i} \geq \alpha \right\}, \quad (6)$$

evaluated at $\alpha \in \{0.90, 0.95, 0.99\}$. We also report the *Explained Variance Ratio* of the first principal component, $\text{EVR}_1 = \lambda_1 / \sum_i \lambda_i$.

Participation Ratio.

$$\text{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \quad (7)$$

High PR signals uniform variance distribution; low PR signals dominance of few directions.

Effective Rank.

$$\text{EffRank} = \exp\left(-\sum_i p_i \log p_i \right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \quad (8)$$

which can be interpreted as the exponential of the spectral entropy. Unlike PCA thresholds, it provides a smooth, threshold-free notion of dimensionality, capturing how uniformly variance is distributed across components.

1.3.2. Nonlinear Estimators

TwoNN.

$$d_{\mathrm{TwoNN}} = \left(\frac{1}{n} \sum_{i=1}^{n} \log \frac{r_2^{(i)}}{r_1^{(i)}} \right)^{-1}, \tag{9}$$

where $r_1^{(i)}, r_2^{(i)}$ are Euclidean distances to the first and second nearest neighbours of $\mathbf{r}_{\ell,b,h}^{(i)}$ (**?**).

MLE.

$$\hat{d}_{\mathrm{MLE}}(\mathbf{r}) = \left(\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{r})}{r_j(\mathbf{r})} \right)^{-1}, \tag{10}$$

averaged over all samples (**?**), with $k = 20$. This estimator is based on a local maximum likelihood principle, assuming uniform sampling in small neighbourhoods, and is sensitive to fine-scale manifold structure beyond global linear approximations.

Linear-Nonlinear (L/N) Ratio

$$\mathrm{Ratio}_B = \frac{\bar{d}_{\mathrm{PCA}_{99}}}{\bar{d}_{\mathrm{TwoNN}}}. \tag{11}$$

Values near 1 indicate near-linear manifolds; higher values signal nonlinear curvature beyond what PCA captures, and serve as a diagnostic for selecting layer targets for spectral reweighting.

1.4. ResiDual Spectral Reweighting

ResiDual applies *spectral reweighting* to selected attention heads of the HTS-AT encoder, modifying how variance is distributed across principal directions before the representation reaches the CLAP projection head. The intervention adds no new learned weights: it relies solely on a PCA basis and scalar reweighting coefficients estimated from unlabelled audio data.

Spectral reweighting operator. Let $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ be the raw output of head h at block (ℓ, b), as defined in Eq. (2). Let $\{\mathbf{u}_j\}_{j=1}^{K}$ be the top-K principal components estimated from the token-level distribution of $\mathbf{H}_{\ell,b,h}$ across the fitting dataset (see fitting procedure below), with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_K$ and empirical mean $\boldsymbol{\mu} \in \mathbb{R}^{d_h}$. ResiDual transforms each token vector $\mathbf{x} = \mathbf{H}_{\ell,b,h}[i,j,:] \in \mathbb{R}^{d_h}$ as:

$$\mathrm{ResiDual}(\mathbf{x}) = \sum_{j=1}^{K} w_j \langle \mathbf{x} - \boldsymbol{\mu},\, \mathbf{u}_j \rangle\, \mathbf{u}_j + \left(\mathbf{x} - \boldsymbol{\mu} - \sum_{j=1}^{K} \langle \mathbf{x} - \boldsymbol{\mu}, \mathbf{u}_j \rangle\, \mathbf{u}_j \right) \tag{12}$$

where $w_j = \lambda_j^{-\gamma/2}$ is the whitening weight for the j-th principal component, controlled by the strength parameter $\gamma \geq 0$. At $\gamma = 0$ the operator is the identity; at $\gamma = 1$ it reduces to full PCA whitening. The out-of-subspace residual (second term) is preserved unchanged, ensuring that directions not captured by the top-K components are not discarded.

Target head selection. The L/N ratio of Eq. (11) serves as a per-head diagnostic: heads with values near 1 exhibit near-linear manifold structure, making their principal subspace well-captured by PCA. Reweighting is applied only to heads in target stages $\ell \in \mathcal{L}$, selected based on this criterion. Hyperparameter choices (K, γ, $\mathcal{L}$) are detailed in Appendix A.4.

Integration into the forward pass. At inference, the `WindowAttention` module of each target block (ℓ, b) is replaced by a `WindowAttentionReweighting` module (Appendix A.4). After computing $\mathbf{H}_{\ell,b,h} = \mathrm{Softmax}(\cdot)\,\mathbf{V}_h$ (Eq. (2)), each head output is independently transformed by $\mathrm{ResiDual}(\cdot)$ before the heads are concatenated and passed through the output projection $W_{\ell,b}^{O}$ (Eq. (19)). The rest of the forward pass is unchanged.

Fitting procedure. The PCA basis for each target head is estimated in a single pass over an unlabelled audio dataset. For each batch, the raw head output $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{B_\times \times M \times d_h}$ — where $B_\times = B \cdot N_w^\ell$ is the effective batch size after window partitioning — is flattened to $(B_\times \cdot M,\, d_h)$ and accumulated in memory. After processing up to $N_{\max}$ audio samples, the accumulated tensors for each head are concatenated into the fitting matrix $\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\mathrm{fit}} \times d_h}$, with $N_{\mathrm{fit}} = \sum_{\mathrm{batches}} B_\times \cdot M$, and standard PCA is applied. The resulting components and eigenvalues are stored as non-trainable buffers. No gradient computation is required.

A. CLAP and HTS-AT: Architecture, Notation, and Residual Decomposition

This appendix is a self-contained reference for the two components at the core of our system. We begin by describing the CLAP dual-encoder model and its configuration (§A.1). We then present the HTS-AT audio backbone in detail, covering input preprocessing, hierarchical stage structure, and the internal mechanics of each attention block (§A.2). We then derive the per-head residual decomposition that underlies our analysis (§A.3). Finally, we provide implementation details for the ResiDual spectral reweighting (§A.4). All notation is introduced inline and collected in Table 3 for reference.

A.1. CLAP Overview

CLAP (Contrastive Language–Audio Pretraining) (**?**) is a dual-encoder model that aligns audio and text representations in a shared embedding space via contrastive learning. The audio encoder is HTS-AT (**?**) and the text encoder is GPT-2 (**?**) (embedding dimension 768), whose weights are frozen throughout training. Both encoders are independently projected to a joint space of dimension $d_{\text{proj}} = 1024$ via dedicated linear projection heads; similarity is measured by temperature-scaled cosine similarity with InfoNCE loss at temperature $\tau = 0.003$. Zero-shot audio classification is performed by comparing an audio embedding against the embeddings of textual class prompts.

The full configuration used in this work is reported in Table 1.

Table 1. CLAP configuration parameters.

Parameter	Value
Text encoder	GPT-2
Text encoder embedding dim	768
Audio encoder	HTS-AT
Audio encoder output dim (D_3)	768
Joint projection dim (d_{proj})	1024
Contrastive temperature τ	0.003
Sampling rate	44 100 Hz
Audio duration	7 s
Mel bands	64
FFT window	1024
Hop size	320
$f_{\min}$ / $f_{\max}$	50 Hz / 8 000 Hz

A.2. HTS-AT Architecture

HTS-AT (**?**) is a Swin-Transformer variant adapted for audio spectrograms. Its computation is organised into four hierarchical *stages*, each composed of several *blocks*, as illustrated in Figure 1. We describe the pipeline from raw audio to the final embedding in order: input preprocessing, patch embedding, stage structure, and the attention mechanism inside each block.

Input preprocessing and patch embedding. Each audio clip is converted to a 64-band log-mel spectrogram ($f_{\min} = 50$ Hz, $f_{\max} = 8{,}000$ Hz, STFT window 1024, hop 320) and normalised per mel-band. The resulting time–frequency matrix is rearranged into a square image $\mathbf{x}_{\text{img}} \in \mathbb{R}^{1\times 256\times 256}$ by folding the time axis into four contiguous segments of 256 frames stacked vertically over the 64 mel-frequency bands ($4\times 64 = 256$ rows, 256 columns). This image is then split into 4×4 non-overlapping patches via a strided `Conv2d` layer (kernel 4×4, stride 4), followed by `LayerNorm`, yielding the input token sequence

$$\mathbf{Z}^{(0,0)} = \mathbf{X}_{\text{in}} \in \mathbb{R}^{4096\times 96}, \tag{13}$$

where $4096 = 64 \times 64$ tokens each have embedding dimension $D_0 = 96$.

Stage structure and PatchMerging. The four stages operate at progressively coarser spatial resolutions, as summarised in Table 2. We index the residual stream as $\mathbf{Z}^{(\ell,b)}$, where $\ell \in \{0, 1, 2, 3\}$ is the stage index and $b \in \{1, \ldots, B_\ell\}$ is the block index within that stage.

After Stages 0, 1, and 2, a *PatchMerging* layer concatenates each 2×2 neighbourhood of spatially adjacent tokens into a single vector of dimension $4D_\ell$, then projects it to $2D_\ell$ via a bias-free linear layer. This halves the spatial side length S_ℓ and doubles the embedding dimension:

$$\mathbb{R}^{S_\ell^2\times D_\ell} \xrightarrow{\text{PatchMerging}} \mathbb{R}^{(S_\ell/2)^2\times 2D_\ell}. \tag{14}$$

Stage 3 has no PatchMerging. The resulting spatial side lengths are $S_\ell \in \{64, 32, 16, 8\}$ and the corresponding embedding dimensions are $D_\ell \in \{96, 192, 384, 768\}$ (see Table 2).

Block computation: W-MSA and MLP. Each block b at stage ℓ applies two sub-layers in sequence, both preceded by LayerNorm and connected via residual additions:

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b-1)} + \text{W-MSA}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b-1)}\Big)\Big), \tag{15}$$

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b)} + \text{MLP}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b)}\Big)\Big). \tag{16}$$

Even-indexed blocks use standard Window Multi-head Self-Attention (W-MSA); odd-indexed blocks use Shifted-Window MSA (SW-MSA), which shifts the partition by $(w/2, w/2)$ tokens to enable cross-window interactions. At Stage 3, where $S_3 = w = 8$ so the grid coincides with a single window, the shift degenerates to zero and all blocks use W-MSA. The MLP sub-layer is a two-layer feed-forward network with hidden dimension $4D_\ell$ and GELU activation.

WindowAttention in detail. The W-MSA module at block (ℓ, b) first partitions the S_ℓ^2 tokens into $N_w^\ell = S_\ell^2/M$ non-overlapping windows of $M = 64$ tokens each, then applies multi-head attention independently within each window.

In the SW-MSA variant, a cyclic shift is applied before window partitioning.

Queries, keys, and values are computed via a single fused

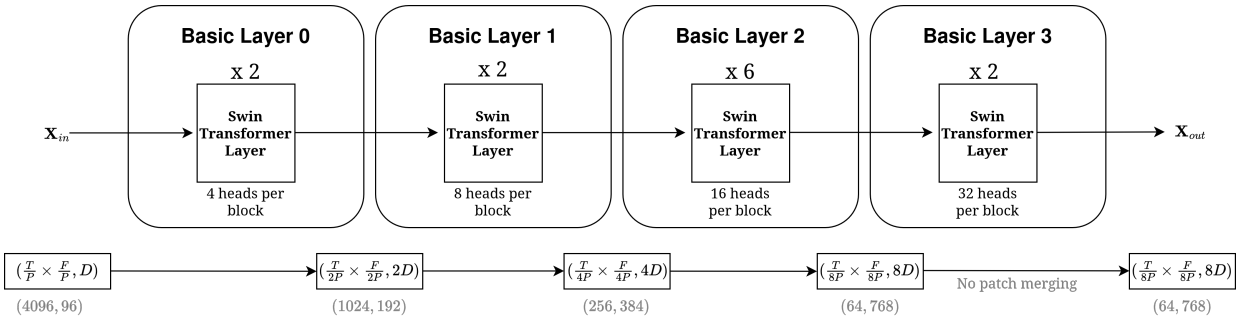

Figure 1. HTS-AT hierarchical architecture. The model consists of four stages of increasing embedding dimension: Stage 0 (2 blocks, 4 heads), Stage 1 (2 blocks, 8 heads), Stage 2 (6 blocks, 16 heads), and Stage 3 (2 blocks, 32 heads). PatchMerging is applied after Stages 0, 1, and 2, halving the spatial side length and doubling the feature dimension at each transition. Stage 3 has no PatchMerging. The input token sequence has spatial size $64 \times 64 = 4{,}096$ tokens with embedding dimension $D_0 = 96$; after three PatchMerging operations the final spatial size is $8 \times 8 = 64$ tokens with $D_3 = 768$.

Table 2. HTS-AT stage-level architectural parameters. The per-head dimension $d_h = D_\ell / H_\ell = 24$ is constant across all stages. S_ℓ is the spatial side length of the token grid at stage ℓ; $N_w^\ell = S_\ell^2 / M$ is the number of attention windows, with $M = w^2 = 64$ tokens per window ($w = 8$).

Stage ℓ	**Blocks** B_ℓ	**Heads** H_ℓ	**Dim** D_ℓ	**Spatial** S_ℓ^2	**Windows** N_w^ℓ
0	2	4	96	64×64	64
1	2	8	192	32×32	16
2	6	16	384	16×16	4
3	2	32	768	8×8	1
Total heads H_{tot}		$2{\cdot}4 + 2{\cdot}8 + 6{\cdot}16 + 2{\cdot}32 = 8 + 16 + 96 + 64 = 184$			

projection $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$:

$$\mathrm{LN}(\mathbf{Z}^{(\ell,b-1)})\, W_{\ell,b}^{QKV} \xrightarrow{\text{split}(D_\ell)} \mathbf{Q},\, \mathbf{K},\, \mathbf{V} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \tag{17}$$

Each tensor is then reshaped to isolate the H_ℓ heads, yielding per-head slices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ with $d_h = D_\ell / H_\ell = 24$.

For each head $h \in \{1, \ldots, H_\ell\}$, attention is computed as:

$$\mathbf{H}_{\ell,b,h} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h} \right) \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{18}$$

where $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head h of block (ℓ, b). This bias encodes the relative spatial offset between each pair of tokens *within* a window; it is shared across all N_w^ℓ windows (broadcast) and is independent for each head. Concretely, it is read from a parameter table of shape $((2w-1)^2, H_\ell) = (225, H_\ell)$ via a fixed index mapping, so each head h has its own column of learnable values.

All H_ℓ head outputs are then concatenated and passed through the block-specific output projection $W_{\ell,b}^O \in \mathbb{R}^{D_\ell \times D_\ell}$ with bias $\mathbf{b}_{\ell,b}^O \in \mathbb{R}^{D_\ell}$, giving the W-MSA output

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \in \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}, \tag{19}$$

which is added to the residual stream in Eq. (15). For a complete step-by-step account of all tensor shapes across every stage, see https://github.com/lorenzo-arcioni/ResiDual-CLAP/blob/main/README.md.

A.3. Per-Head Residual Decomposition

The additive residual structure of Eq. (15) allows us to decompose the W-MSA contribution $\mathbf{A}_{\ell,b}$ exactly into independent per-head terms. Denoting by $W_{\ell,b,h}^O \in \mathbb{R}^{d_h \times D_\ell}$ the row slice of $W_{\ell,b}^O$ corresponding to head h (rows $[(h-1)d_h,\ hd_h)$), we distribute the output projection over heads:

$$\begin{aligned} \mathbf{A}_{\ell,b} &= \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \\ &= \sum_{h=1}^{H_\ell} \left(\mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \frac{\mathbf{b}_{\ell,b}^O}{H_\ell} \right) = \sum_{h=1}^{H_\ell} \widehat{\mathbf{H}}_{\ell,b,h}, \end{aligned} \tag{20}$$

where we define the *per-head projected contribution*

$$\widehat{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h}\, W^{O}_{\ell,b,h} + \frac{\mathbf{b}^{O}_{\ell,b}}{H_\ell} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times D_\ell}. \quad (21)$$

This decomposition is exact and follows from the linearity of matrix multiplication: because $W^{O}_{\ell,b}$ acts on the concatenation of all head outputs, each head h effectively multiplies only its own row slice $W^{O}_{\ell,b,h}$, and the output bias can be distributed equally among heads without any approximation.

The decomposition holds within a single stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension, breaking any global additive structure; cross-stage comparisons must therefore account for this.

A.4. ResiDual: Implementation Details

Module injection. For each target block (ℓ, b) with $\ell \in \mathcal{L}$, the standard Window Multi-head Self-Attention is replaced in-place by the reweighted variant. All pretrained parameters — the fused QKV projection $W^{QKV}_{\ell,b}$, the output projection $W^{O}_{\ell,b}$ with bias $\mathbf{b}^{O}_{\ell,b}$, and the relative position bias table from which $\mathbf{B}_{\ell,b,h}$ is read — are copied exactly from the original module. The only additions are non-trainable PCA buffers, one set per head (described below). A shared accumulation dictionary, indexed by (ℓ, b, h), coordinates the collection of head activations across all target blocks during the fitting pass.

Per-head spectral layer. Each head $h \in \{1, \dots, H_\ell\}$ of each target block (ℓ, b) with $\ell \in \mathcal{L}$ is equipped with an independent spectral reweighting layer operating in $\mathbb{R}^{d_h}$, where $d_h = 24$ is constant across all stages. The layer retains $K = \lfloor d_h \cdot \rho \rfloor$ principal components, where $\rho \in (0, 1)$ is the components-ratio hyperparameter. It stores three non-trainable buffers: the PCA component matrix $\mathbf{\Phi}^{(K)}_{\ell,b,h} \in \mathbb{R}^{d_h\times K}$ (the top-K eigenvectors of the empirical covariance of $\mathbf{R}_{\ell,b,h}$), the empirical mean $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$, and the reweighting vector $\mathbf{w}_{\ell,b,h} \in \mathbb{R}^{K}$ with j-th entry

$$w_j = \max\!\big(\lambda_j^{-\gamma/2},\; \varepsilon\big), \quad (22)$$

where λ_j is the j-th eigenvalue of the empirical covariance $\mathbf{C}_{\ell,b,h}$, γ is the whitening-strength hyperparameter, and ε is a numerical-stability clamp. Before the PCA basis is fitted the layer acts as the identity map.

Fitting pass. The fitting pass is run with gradients disabled and the model in evaluation mode. For each audio clip, the forward pass is executed with data collection enabled. Each target block intercepts, for every head h, the pre-reweighting tensor $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}$ and flattens it to $\mathbb{R}^{(N^{\ell}_{w}\cdot M)\times d_h}$ before appending it to the accumulation buffer for (ℓ, b, h). After all samples have been processed (up to $N_{\max}$ audio clips), the buffers are concatenated along the first axis to form the fitting matrix

$$\widetilde{\mathbf{R}}_{\ell,b,h} \;\in\; \mathbb{R}^{N_{\text{fit}}\times d_h}, \qquad N_{\text{fit}} \;=\; N_{\max}\cdot N^{\ell}_{w}\cdot M. \quad (23)$$

Standard PCA is then applied: the empirical mean $\boldsymbol{\mu}_{\ell,b,h}$ is computed and subtracted, and the empirical covariance is decomposed as

$$\begin{aligned}\mathbf{C}_{\ell,b,h} &= \frac{1}{N_{\text{fit}}-1}\Big(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h}\Big)^{\top}\Big(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h}\Big)\\ &= \mathbf{\Phi}_{\ell,b,h}\;\mathrm{diag}(\lambda_1,\dots,\lambda_{d_h})\;\mathbf{\Phi}^{\top}_{\ell,b,h}, \quad (24)\\ &\text{such that } \lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_{d_h} \ge 0. \quad (25)\end{aligned}$$

The top-K columns of $\mathbf{\Phi}_{\ell,b,h}$, denoted $\mathbf{\Phi}^{(K)}_{\ell,b,h} \in \mathbb{R}^{d_h\times K}$, and the corresponding eigenvalues $\lambda_1, \dots, \lambda_K$ are stored as non-trainable buffers; the reweighting vector $\mathbf{w}_{\ell,b,h}$ is initialised via Eq. (22). The temporary accumulation buffers are then cleared to free memory.

Forward pass with reweighting. The standard attention computation of Eq. (18) produces, for each head h, the raw head output $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}$. Spectral reweighting is applied as follows. The output is first centred by subtracting the empirical mean,

$$\widetilde{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}, \quad (26)$$

where $\boldsymbol{\mu}_{\ell,b,h}$ is broadcast over the first two axes. The centred tensor is then projected onto the K-dimensional PCA subspace,

$$\mathbf{Z}_{\ell,b,h} \;=\; \widetilde{\mathbf{H}}_{\ell,b,h}\,\mathbf{\Phi}^{(K)}_{\ell,b,h} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times K}, \quad (27)$$

and each principal component is independently rescaled by the corresponding entry of $\mathbf{w}_{\ell,b,h}$,

$$\widehat{\mathbf{Z}}_{\ell,b,h} \;=\; \mathbf{Z}_{\ell,b,h} \odot \mathbf{w}^{\top}_{\ell,b,h} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times K}, \quad (28)$$

where $\mathbf{w}^{\top}_{\ell,b,h}$ is broadcast over the first two axes. The rescaled coordinates are mapped back to $\mathbb{R}^{d_h}$ via the transpose basis,

$$\widehat{\mathbf{H}}^{\text{PCA}}_{\ell,b,h} \;=\; \widehat{\mathbf{Z}}_{\ell,b,h}\,\big(\mathbf{\Phi}^{(K)}_{\ell,b,h}\big)^{\top} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}. \quad (29)$$

The component of $\widetilde{\mathbf{H}}_{\ell,b,h}$ orthogonal to the PCA subspace is retained unchanged,

$$\mathbf{E}_{\ell,b,h} \;=\; \widetilde{\mathbf{H}}_{\ell,b,h} - \mathbf{Z}_{\ell,b,h}\,\big(\mathbf{\Phi}^{(K)}_{\ell,b,h}\big)^{\top} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}, \quad (30)$$

and the reweighted head output is obtained by recombining the two terms and restoring the empirical mean,

$$\mathbf{H}^{*}_{\ell,b,h} \;=\; \widehat{\mathbf{H}}^{\text{PCA}}_{\ell,b,h} + \mathbf{E}_{\ell,b,h} + \boldsymbol{\mu}_{\ell,b,h} \;\in\; \mathbb{R}^{N^{\ell}_{w}\times M\times d_h}. \quad (31)$$

The tensor $\mathbf{H}^{*}_{\ell,b,h}$ is substituted for $\mathbf{H}_{\ell,b,h}$ in Eq. (19), so the output projection $W^{O}_{\ell,b}$ and the residual stream update of Eq. (15) proceed unchanged.

Hyperparameter summary. Table 4 summarises the hyperparameters introduced by ResiDual and the default values used in all experiments.

B. Post-Projection Head Analysis

The main analysis operates on *raw pre-projection* head outputs $\mathbf{H}_{\ell,b,h}$, which characterise the intrinsic computational geometry of each head independently of W^O. Here we describe a complementary analysis of the *projected* contributions $\widehat{\mathbf{H}}_{\ell,b,h}$, which reflect each head's additive influence on the residual stream after the output projection.

B.1. Motivation and Methodological Differences

The per-head projected contribution $\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$ is obtained by applying the row slice $W^O_{\ell,b,h}$ of the output projection to the raw head output, and distributing the output bias equally across heads (see Appendix A.3 for the exact decomposition). It lives in the full residual-stream space $\mathbb{R}^{D_\ell}$ of stage ℓ, offering a complementary analytical perspective:

1. **Residual-stream interpretability.** $\widehat{\mathbf{H}}_{\ell,b,h}$ is the quantity that each head writes additively to $\mathbf{Z}^{(\ell,b)}$ via Eq. (1). Its geometry therefore directly characterises each head's contribution to the representations that are consumed by downstream layers and, ultimately, the CLAP projection head P.
2. **Role of W^O.** Comparing pre- and post-projection dimensionality estimates reveals how the output projection transforms each head's internal computation: if dimensionality increases after projection, W^O expands the head's representational geometry in the residual stream; if it decreases, W^O compresses or rotates it.

B.2. Extraction Procedure

The same forward hooks described in Section 1.2 are reused. After intercepting $\mathbf{H}_{\ell,b,h}$ of shape (N_w^ℓ, M, d_h), we reconstruct the projected contribution by applying the corresponding row slice $W^O_{\ell,b,h} \in \mathbb{R}^{d_h \times D_\ell}$ retrieved from `self.proj.weight`, and adding the distributed bias term $\mathbf{b}^O_{\ell,b}/H_\ell$, as in Eq. (20):

$$\widehat{\mathbf{H}}_{\ell,b,h} = \mathbf{H}_{\ell,b,h}\, W^O_{\ell,b,h} + \frac{\mathbf{b}^O_{\ell,b}}{H_\ell} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (32)$$

We then spatially aggregate by mean-pooling over all windows and token positions:

$$\widehat{\mathbf{r}}_{\ell,b,h} = \frac{1}{N_w^\ell M} \sum_{i=1}^{N_w^\ell} \sum_{j=1}^{M} \widehat{\mathbf{H}}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{D_\ell}, \quad (33)$$

yielding the dataset matrix $\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n \times D_\ell}$ by stacking across n samples.

B.3. Dimensionality Estimators in the Post-Projection Space

All estimators from Section 1.3 apply with $\widehat{\mathbf{R}}_{\ell,b,h}$ replacing $\mathbf{R}_{\ell,b,h}$. The key difference is that the ambient dimension now varies across stages: $D_\ell \in \{96, 192, 384, 768\}$.

PCA. The maximum number of non-trivial principal components of $\widehat{\mathbf{R}}_{\ell,b,h}$ is bounded by $\min(n, D_\ell)$, so a head at Stage 3 ($D_3 = 768$) has a strictly larger ambient ceiling than a head at Stage 0 ($D_0 = 96$). Direct comparison of explained-variance curves or intrinsic dimensionality estimates *across* stages is therefore confounded by this varying ceiling and must account for it explicitly; within a single stage, where D_ℓ is fixed, comparisons across blocks and heads are well-defined.

L/N Ratio. The ratio $d_{\mathrm{PCA}_{99}}/d_{\mathrm{TwoNN}}$ must be interpreted with awareness that the PCA ceiling grows with stage, which may inflate linear dimensionality estimates independently of any genuine increase in representational complexity.

B.4. Interpretation and Relationship to Main Results

Post-projection analysis is best suited to studying *how heads collectively shape the residual stream* and, ultimately, the CLAP embedding used for zero-shot classification. Pre-projection analysis—the perspective adopted in the main body—is best suited to studying *individual head specialisation* in isolation, free from the mixing introduced by W^O.

Concretely, if a head exhibits low intrinsic dimensionality in the pre-projection space but high dimensionality after projection, W^O acts as an expansion map for that head in the residual stream; the converse implies compression or selective routing of the head's computation into a lower-dimensional subspace of $\mathbb{R}^{D_\ell}$.

Limits of cross-stage analysis. A key difference between HTS-AT and isotropic vision transformers such as ViT (**?**) is that the ambient dimension of the residual stream is not constant: $D_\ell \in \{96, 192, 384, 768\}$ grows with each stage. As a consequence, the matrices $\widehat{\mathbf{R}}_{\ell,b,h}$ from different stages live in spaces of different dimension, and any dimensionality-reduction analysis—such as PCA—must be interpreted with care. Specifically, the maximum number of non-trivial principal components of $\widehat{\mathbf{R}}_{\ell,b,h}$ is bounded by $\min(n, D_\ell)$, so a head at Stage 3 ($D_3 = 768$) has a strictly larger ambient ceiling than a head at Stage 0 ($D_0 = 96$). Direct comparison of explained-

variance curves or intrinsic dimensionality estimates across stages is therefore confounded by this varying ceiling, and must account for it explicitly. Within a single stage, where D_ℓ is fixed, comparisons across blocks and heads are well-defined and unambiguous.

C. Extended Dimensionality Analysis

This appendix provides comprehensive quantitative details and additional visualizations complementing the main results in Section **??**.

C.1. Detailed Block-Level Statistics

Table 6 reports complete block-wise metrics across all 12 transformer blocks in HTS-AT, aggregating over the heads within each block as described in Section 1.3.

Relationship to Architecture. As illustrated in Figure 1, the spatial resolution decreases progressively through the network due to patch merging between stages. While this affects the number of tokens N processed by each attention head, our analysis focuses on the intrinsic dimensionality of the *head dimension* $d_h = 24$ after spatial aggregation (Eq. 4). Thus, the reported metrics characterize the semantic complexity of head representations independent of spatial resolution effects.

The hierarchical structure creates natural breakpoints for dimensionality analysis:

- **Stage 1 (Blocks 0–1)**: High spatial resolution ($T/2P \times F/2P$) but limited capacity ($D_0 = 96$). Early fusion of local spectral-temporal patterns.
- **Stage 2 (Blocks 2–3)**: First dimensionality jump coincides with 2× patch merging and head doubling. Transition from local to intermediate-scale features.
- **Stage 3 (Blocks 4–9)**: Deepest stage with 6 blocks enables iterative refinement at fixed spatial scale ($T/4P \times F/4P$) and capacity ($D_2 = 384$). Gradual dimensionality growth reflects progressive feature abstraction.
- **Stage 4 (Blocks 10–11)**: Maximum capacity ($D_3 = 768$) without further spatial reduction. Minimal dimensionality increase suggests saturation.

Interpretation.

- Block 0 operates near the linear regime (L/N $\approx$ 1), with almost 88% variance in the first PC, indicating extremely constrained early processing.
- The largest single-block jump occurs at the Stage 1→2 transition (blocks 1→2: ΔL = +4.75, +59%), corresponding to doubling of attention heads (4→8) and hidden dimension (96→192).
- Stage 3 exhibits gradual linear ID growth (18.00 → 22.25 over 6 blocks) despite constant architecture, suggesting intra-stage feature refinement through depth.
- Stage 4 shows minimal progression (blocks 10→11: ΔL = -0.66), consistent with representational saturation observed in the main text.

C.2. Extended Cross-Dataset Analysis

We replicate the full layer-wise analysis on ESC-50 and VocalSound to validate architectural generalizability. Tables 7 and 8 present complete statistics.

Cross-Dataset Consistency Analysis. Despite differing semantic granularities (ESC-50: 50 classes, VocalSound: 6 classes, TinySOL: 14 classes), layer-wise trends remain remarkably stable:

- **L0 Concentration**: All datasets exhibit EVR1 ¿ 69% in Stage 1, confirming universal early spectral concentration.
- **L1 Expansion**: The L0→L1 dimensionality jump is consistent (TinySOL: +9.2, ESC-50: +9.1, VocalSound: +10.1 for $d_{\text{PCA}_{99}}$), with coefficient of variation across datasets CV = 0.06.
- **L2-L3 Saturation**: All datasets show similar modest L2→L3 growth ($\Delta d < 2.0$), despite L3 having 2× the heads of L2, indicating architecture-driven capacity limits.
- **L/N Ratio Convergence**: By Stage 4, all datasets reach L/N $\approx$ 2.6–2.7, suggesting a universal nonlinear complexity regime independent of semantic domain.

C.3. Statistical Validation

C.3.1. ANOVA Results

One-way ANOVA tests for layer differences on TinySOL dataset:

C.3.2. Post-Hoc Pairwise Comparisons

Bonferroni-corrected pairwise t-tests for $d_{\text{PCA}_{99}}$ (6 comparisons, $\alpha_{\text{corrected}} = 0.0083$):

All effect sizes exceed Cohen's threshold for "large" effects ($d > 0.8$), with the L0 vs L3 comparison exhibiting extremely large effects ($d > 8$), confirming substantial representational differences across layers.

C.4. Additional Visualizations

C.4.1. PC1 Dominance and Boxplots

Figure 2 presents complementary views of dimensionality structure.

Observations from Panel (a):

- Only 2 heads in L0 (2.1%) fall below the 50% EVR1 threshold, compared to 89% of L3 heads, demonstrating near-universal early concentration.
- The EVR1 distribution shifts from unimodal (L0: concentrated near 0.7–0.8) to bimodal (L3: peaks at 0.2–0.3 and 0.35–0.4), suggesting emergence of head subpopulations with distinct specialization levels.

Observations from Panel (b):

- PR and EffRank exhibit parallel scaling, confirming their measurement of related spectral properties.
- TwoNN shows compressed scale relative to PCA99, visually emphasizing the linear-nonlinear gap discussed in the main text.
- Outliers (marked as individual points beyond whiskers) are rare in L0-L1 but frequent in L2-L3, consistent with increased head-level heterogeneity.

C.5. Computational Details

All analyses were performed on an NVIDIA A100 GPU (40GB) using PyTorch 2.0.1 and Python 3.10. Key implementation details:

- **Head Extraction**: Forward hooks registered via `torch.nn.Module.register_forward_hook`. Batch size 100 for extraction to balance memory and throughput.
- **PCA**: Computed via `sklearn.decomposition.PCA` with full SVD solver. Eigenvalue thresholding at machine epsilon ($\sim 10^{-7}$) to remove numerical noise.
- **TwoNN**: Implemented using `skdim.id.TwoNN` with default parameters (no k selection required).
- **MLE**: `skdim.id.MLE` with $k = 20$ neighbors, standard Euclidean metric.
- **Statistical Tests**: `scipy.stats` functions (`f_oneway`, `ttest_ind`, `spearmanr`) with standard settings.

Total extraction time: $\sim$45 minutes per dataset (1000 samples $\times$ 184 heads). Analysis pipeline code available at https://github.com/[ANONYMOUS]/residual-audio.

C.6. Reproducibility Checklist

To facilitate replication:

- Random seeds: 42 (Python), 42 (NumPy), 42 (PyTorch)
- CLAP version: `laion/clap-htsat-unfused` checkpoint from HuggingFace
- Audio preprocessing: CLAP default (64-band mel, 10s duration, 48kHz resampling)
- Dataset versions: ESC-50 v2.0, TinySOL v3.0, VocalSound official release
- Stratified sampling: `sklearn.model_selection.StratifiedShuffleSplit` with 1000 samples

panel_E_pc1_dominance.png

(a) PC1 Variance Dominance

panel_F_boxplot.png

(b) Multi-Metric Boxplots

Figure 2. Extended dimensionality analysis. (a) First principal component variance explained across all 184 heads. Horizontal line at 50% marks equal-contribution threshold. Sharp decline from L0 (mean 73%) to L3 (mean 27%) quantifies transition from low-rank to distributed representations. (b) Boxplot comparison of four key metrics across layers, revealing consistent monotonic trends and increasing intra-layer variance in deeper stages (note wider boxes for L2-L3).

Table 3. Summary of notation used throughout the paper.

Symbol	Definition	Values / Notes
$\ell \in \{0,1,2,3\}$	Stage index	
$b \in \{1,\ldots,B_\ell\}$	Block index within stage ℓ	$B_\ell \in \{2,2,6,2\}$
$h \in \{1,\ldots,H_\ell\}$	Attention head index within block (ℓ,b)	$H_\ell = 4 \cdot 2^\ell \in \{4,8,16,32\}$
S_ℓ	Spatial side length of the token grid at stage ℓ	$S_\ell \in \{64,32,16,8\}$
$w = 8$	Attention window side length	
$M = w^2 = 64$	Tokens per attention window	
$N_w^\ell = S_\ell^2/M$	Number of attention windows at stage ℓ	$N_w^\ell \in \{64,16,4,1\}$
$d_h = 24$	Per-head feature dimension (constant across stages)	$d_h = D_\ell/H_\ell$
$D_\ell = H_\ell \cdot d_h$	Total embedding dimension at stage ℓ	$D_\ell \in \{96,192,384,768\}$
$d_{\text{proj}} = 1024$	CLAP joint embedding dimension	
$W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$	Fused QKV projection at block (ℓ,b)	
$W_{\ell,b}^{O} \in \mathbb{R}^{D_\ell \times D_\ell}$	Output projection at block (ℓ,b)	
$W_{\ell,b,h}^{O} \in \mathbb{R}^{d_h \times D_\ell}$	Row slice of $W_{\ell,b}^O$ for head h	rows $[(h-1)d_h,\ hd_h)$
$\mathbf{b}_{\ell,b}^{O} \in \mathbb{R}^{D_\ell}$	Output projection bias at block (ℓ,b)	
$\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$	Relative position bias for head h at block (ℓ,b)	
$\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Raw head output at block (ℓ,b), head h (pre-$W_{\ell,b}^O$)	
$\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$	Per-head projected contribution (post-$W_{\ell,b}^O$); see Eq. (21)	
$\mathbf{r}_{\ell,b,h} \in \mathbb{R}^{d_h}$	Spatially aggregated raw head output; see Eq. (4)	
$\widehat{\mathbf{r}}_{\ell,b,h} \in \mathbb{R}^{D_\ell}$	Spatially aggregated projected head output; see Eq. (33)	
$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$	Dataset matrix of $\mathbf{r}_{\ell,b,h}$ across n samples	
$\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n \times D_\ell}$	Dataset matrix of $\widehat{\mathbf{r}}_{\ell,b,h}$ across n samples	
$P : \mathbb{R}^{768} \to \mathbb{R}^{1024}$	CLAP audio projection head (two-layer MLP with residual)	
n	Number of audio samples in the dataset	
$\mathcal{L} \subseteq \{0,1,2,3\}$	Target stage indices for reweighting	Default $\{2,3\}$
$K = \lfloor d_h \cdot \rho \rfloor$	Number of retained PCA components per head	
$\rho \in (0,1)$	PCA components ratio	Default 0.25
$\gamma \geq 0$	Whitening strength	Default 1.0
$\varepsilon > 0$	Numerical stability clamp for reweighting coefficients	Default 10^{-6}
$N_{\max}$	Maximum number of audio clips used in fitting pass	Default 10 000
N_{fit}	Total rows of the fitting matrix; $N_{\text{fit}} = \sum_{\text{batches}} N_w^\ell \cdot M \cdot B$	
$\mathbf{C}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Empirical covariance matrix of $\widetilde{\mathbf{R}}_{\ell,b,h}$	See Eq. (25)
$\mathbf{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Orthogonal matrix of eigenvectors of $\mathbf{C}_{\ell,b,h}$	Columns ordered by $\lambda_1 \geq \cdots \geq \lambda_{d_h}$
$\mathbf{\Phi}_{\ell,b,h}^{(K)} \in \mathbb{R}^{d_h \times K}$	First K columns of $\mathbf{\Phi}_{\ell,b,h}$ (top-K PCA basis)	
λ_j	j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$, $\lambda_1 \geq \cdots \geq \lambda_{d_h} \geq 0$	
$\mathbf{w}_{\ell,b,h} \in \mathbb{R}^K$	Reweighting vector; $w_j = \max(\lambda_j^{-\gamma/2}, \varepsilon)$	
$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}} \times d_h}$	Fitting matrix for head h of block (ℓ,b)	See Eq. (23)
$\widetilde{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Centred head output ($\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}$)	
$\mathbf{Z}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times K}$	Projection of $\widetilde{\mathbf{H}}_{\ell,b,h}$ onto top-K PCA components	
$\widehat{\mathbf{Z}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times K}$	Rescaled PCA projection ($\mathbf{Z}_{\ell,b,h} \odot \mathbf{w}_{\ell,b,h}^\top$)	
$\widehat{\mathbf{H}}_{\ell,b,h}^{\text{PCA}} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Reconstruction of $\widetilde{\mathbf{H}}_{\ell,b,h}$ in the PCA subspace after rescaling	
$\mathbf{E}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Orthogonal residual: component of $\widetilde{\mathbf{H}}_{\ell,b,h}$ outside the PCA subspace	
$\mathbf{H}_{\ell,b,h}^{*} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Reweighted head output; replaces $\mathbf{H}_{\ell,b,h}$ in Eq. (19)	

Table 4. ResiDual hyperparameters.

Symbol	Description	Default
$\mathcal{L}$	Target stage indices for reweighting	$\{2, 3\}$
ρ	PCA components ratio ($K = \lfloor d_h \cdot \rho \rfloor$)	0.25
γ	Whitening strength (see Eq. (22))	1.0
ε	Numerical stability clamp for w_j	10^{-6}
$N_{\max}$	Maximum fitting samples	$10\,000$

Table 5. Comparison of the two analysis pipelines.

	Main (pre-W^O)	**Appendix (post-W^O)**
Representation	$\mathbf{H}_{\ell,b,h}$	$\widehat{\mathbf{H}}_{\ell,b,h}$
Ambient dim	$d_h = 24$ (constant)	$D_\ell \in \{96, 192, 384, 768\}$
Analysis space	$\mathbb{R}^{24}$ (native)	$\mathbb{R}^{D_\ell}$
Hook location	Before `.reshape` only	Before `.reshape`, then $\times W_h^O$
W^O applied	No	Yes (via W_h^O)
P applied	No	No
Captures	Internal attention computation	Contribution to residual stream

Table 6. Block-wise aggregated dimensionality metrics for TinySOL dataset. Blocks 0–1 (Stage 1), 2–3 (Stage 2), 4–9 (Stage 3), 10–11 (Stage 4). L = Linear ID ($d_{\text{PCA}_{99}}$), N = Nonlinear ID (TwoNN), L/N = Linear-nonlinear ratio, EVR1 = First PC variance explained.

Block	Stage	Heads	L	N	L/N	EVR1
0	1	4	3.75	3.94	0.95	0.878
1	1	4	8.00	4.94	1.62	0.575
2	2	8	12.75	6.15	2.07	0.403
3	2	8	17.50	6.75	2.59	0.460
4	3	16	18.00	6.93	2.60	0.388
5	3	16	20.13	7.08	2.84	0.279
6	3	16	21.25	7.40	2.87	0.250
7	3	16	21.38	7.42	2.88	0.243
8	3	16	22.25	8.37	2.66	0.217
9	3	16	21.88	8.11	2.70	0.241
10	4	32	22.25	8.49	2.62	0.262
11	4	32	21.59	8.00	2.70	0.272

Table 7. Layer-wise dimensionality metrics for ESC-50 dataset (50 environmental sound classes, 1000 stratified samples).

Layer	$d_{\text{PCA}_{99}}$	**TwoNN**	**PR**	**EVR1**
L0	5.2 ± 2.1	4.2 ± 0.7	1.8 ± 0.8	0.741 ± 0.187
L1	14.3 ± 2.8	6.2 ± 0.6	3.9 ± 1.3	0.449 ± 0.109
L2	20.1 ± 2.0	7.4 ± 0.8	7.5 ± 1.8	0.281 ± 0.081
L3	20.3 ± 1.2	7.8 ± 1.1	7.4 ± 1.9	0.279 ± 0.074

Table 8. Layer-wise dimensionality metrics for VocalSound dataset (6 vocal sound categories, 1000 stratified samples).

Layer	$d_{\text{PCA}_{99}}$	**TwoNN**	**PR**	**EVR1**
L0	6.1 ± 2.3	4.6 ± 0.8	2.1 ± 1.0	0.698 ± 0.192
L1	16.2 ± 2.4	6.7 ± 0.5	4.5 ± 1.5	0.421 ± 0.117
L2	21.9 ± 1.9	8.0 ± 0.9	8.2 ± 2.0	0.263 ± 0.079
L3	22.7 ± 1.1	8.6 ± 0.8	8.3 ± 1.6	0.254 ± 0.071

Table 9. Statistical significance of layer effects on dimensionality metrics (TinySOL, $n = 184$ heads). All tests use $\alpha = 0.05$.

Metric	**F-statistic**	**p-value**	**Significance**
$d_{\text{PCA}_{99}}$	262.64	< 0.001	***
TwoNN	58.93	< 0.001	***
PR	44.74	< 0.001	***
EffRank	76.03	< 0.001	***
EVR(PC1)	118.47	< 0.001	***

Table 10. Pairwise layer comparisons for linear intrinsic dimensionality (TinySOL). All comparisons significant at corrected $\alpha = 0.0083$.

Comparison	$\Delta d_{\text{PCA}_{99}}$	**Cohen's** d	**p-value**
L0 vs L1	9.22	3.86	< 0.001
L0 vs L2	14.93	7.12	< 0.001
L0 vs L3	16.04	8.94	< 0.001
L1 vs L2	5.71	2.34	< 0.001
L1 vs L3	6.82	3.19	< 0.001
L2 vs L3	1.11	0.78	< 0.001